Points to remember:

- A way of designing algorithm is called algorithmic strategy.
- A space-time or time-memory tradeoff is a way of solving a problem or calculation in less time by using more storage space.
- Asymptotic Notations are languages that uses meaningful statements about time and space complexity.
- Bubble sort is a simple sorting algorithm. It compares each pair of adjacent items and swaps them if they are in the unsorted order.
- The selection sort improves on the bubble sort by making only one exchange for every pass through the list.
- Insertion sort is a simple sorting algorithm that builds the final sorted array or list one item at a time. It always maintains a sorted sublist in the lower positions of the list.
- Dynamic programming is used when the solutions to a problem can be viewed as the result of a sequence of decisions.

Evaluation

9DJFRN

Part - I

Choose the best answer: **(1 Marks)**

1. The word comes from the name of a Persian mathematician Abu Ja'far Mohammed ibn-i Musa al Khowarizmi is called?

   (A) Flowchart (B) Flow (C) Algorithm (D) Syntax

2. From the following sorting algorithms which algorithm needs the minimum number of swaps?

   (A) Bubble sort (B) Quick sort (C) Merge sort (D) Selection sort

3. Two main measures for the efficiency of an algorithm are

   (A) Processor and memory (B) Complexity and capacity

   (C) Time and space (D) Data and space

4. The complexity of linear search algorithm is

   (A) O(n) (B) O(log n) (C) O(n2) (D) O(n log n)

5. From the following sorting algorithms which has the lowest worst case complexity?

   (A) Bubble sort (B) Quick sort (C) Merge sort (D) Selection sort